

Programmazione Concorrente e Parallela e su Cloud
Prof. Vittorio Scarano



Università degli Studi di Salerno

Amazon Web Services:
Introduction

Carmine Spagnuolo
2018

Outline

Cluster Architecture

Cloud Computing

Virtualization Technologies

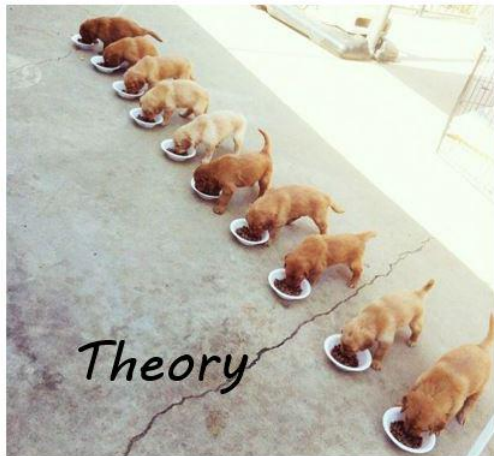
Cloud Computing Platform

Amazon Web Services (AWS)

Benchmarking

The lesson assumption

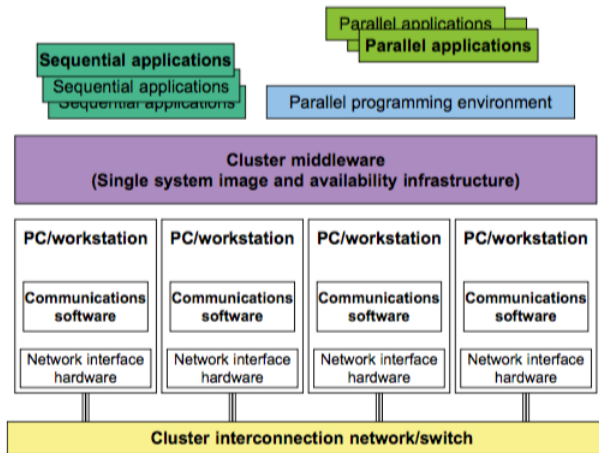
Multithreaded programming



Cluster Architecture

Cluster Architecture

Architecture of a computer cluster built with commodity even heterogeneous hardware, software, middleware, and network.



Cluster Architecture

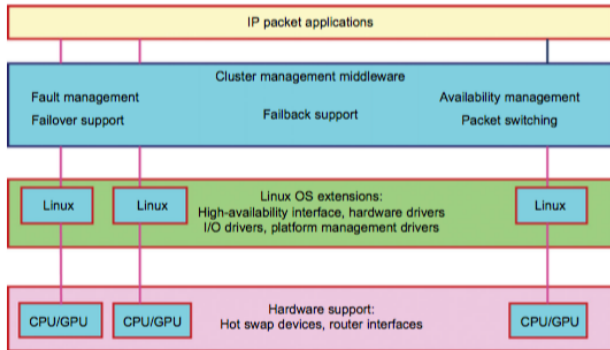
Dedicated hardware is an instance High Performance Computing typically named Supercomputing.

Table 2.2 The Top Five Supercomputers Evaluated in November 2010

System Name, Site, and URL	System Name, Processors, OS, Topology, and Developer	Linpack Speed (R_{max}), Power	Efficiency (R_{max}/R_{peak})
1. Tianhe-1A, National Supercomputing Center, Tianjin, China, http://www.nscn-tj.gov.cn/en/	NUDT TH1A with 14,336 Xeon X5670 CPUs (six cores each) plus 7168 NVIDIA Tesla M2050 GPUs (448 CUDA cores each), running Linux, built by National Univ. of Defense Technology, China	2.57 Pflops, 4.02 MW	54.6% (over a peak of 4.7 Pflops)
2. Jaguar, DOE/SC/Oak Ridge National Lab., United States, http://computing.ornl.gov	Cray XT5-HE: MPP with 224,162 x 6 AMD Opteron, 3D torus network, Linux (CLE), manufactured by Cray, Inc.	1.76 Pflops, 6.95 MW	75.6% (over a peak of 4.7 Pflops)
3. Nebulae at China's National Supercomputer Center, ShenZhen, China http://www.ict.cas.cas.cn	TC3600 Blade, 120,640 cores in 55,680 Xeon X5650 plus 64,960 NVIDIA Tesla C2050 GPUs, Linux, InfiniBand, built by Dawning, Inc.	1.27 Pflops, 2.55 MW	42.6% (over a peak of 2.98 Pflops)
4. TSUBAME 2.0, GSIC Center, Tokyo Institute of Technology, Tokyo, Japan, http://www.gsic.titech.ac.jp/	HP cluster, 3000SL, 73,278 x 6 Xeon X5670 processors, NVIDIA GPU, Linux/SLES 11, built by NEC/HP	1.19 Pflops, 1.8 MW	52.19% (over a peak of 2.3 Pflops)
5. Hopper, DOE/SC/LBNL/ NERSC, Berkeley, CA. USA, http://www.nersc.gov/	Cray XE6 150,408 x 12 AMD Opteron, Linux (CLE), built by Cray, Inc.	1.05 Pflops, 2.8 MW	78.47% (over a peak of 1.35 Pflops)

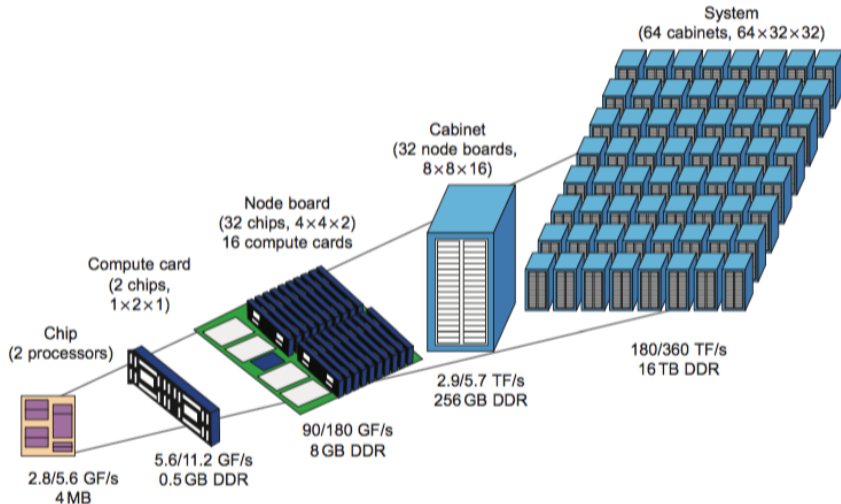
Exploiting Data-parallel computing

Middleware, Linux extensions, and hardware support for achieving massive parallelism and high availability in a Linux cluster system built with CPUs and GPUs.



IBM Blue Gene/L

Architecture built with modular components packaged hierarchically in five levels.



Cluster Architecture



[5] Mira (USA) is IBM's self-built Supercomputer that has the peak performance of 8.6 petaflops. At Argonne National Laboratory, US, it uses IBM's BlueGene/Q platform with 786,000 core processors.

Cluster Architecture



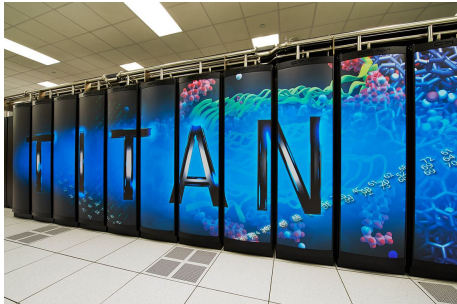
[4] K Computer (Japan) is Fujitsu's K computer is located at Japan's RIKEN Advanced Institute for Computational Science, with 10.5 petaflops. With 705,000 SPARC64 VIIIfx octo-core processors, K computer performs scientific operations including global disaster prevention, meteorology and medical research.

Cluster Architecture



[3] Sequoia (USA) is IBM machine Sequoia also runs on BlueGene/Q design with 1.6 million 16- core processors. It has a performing speed of 17.2 petaflops.

Cluster Architecture



[2] **Titan (USA)** is a major upgrade to another Supercomputer Jaguar, Titan performs at a blazing speed of 17.6 petaflops. It uses a total of 561,000 Opteron 6274 16-core 2.2GHz processors along with NVIDIA GPUs.

Cluster Architecture



[1] Tianhe-2 (China) literally meaning Milky Way-2, the king of supercomputers processes at a staggering speed of 33.9 petaflops. It was developed by China's National University of Defense Technology. This peerless Supercomputer runs on a whopping 3,120,000 cores, comprising of a mix of Intel Xeon E5 processors, custom processors and Intel Xeon Phi coprocessors.

Cluster Architecture



[NEW GENERATION] Aurora (USA) is a revolutionary architecture that exploits Intel's HPC scalable system framework and 2nd generation Intel Omni-Path Fabric and Cray Inc. processors.

The system will have a combined total of over 8 Petabytes of on package high bandwidth memory and persistent memory, connected and communicating via a high-performance system fabric to achieve landmark throughput.

The nodes will be linked to a dedicated burst buffer and a high-performance parallel storage solution.

<http://aurora.alcf.anl.gov/>

Access to a Cluster Machine

A cluster of servers interconnected by a high-bandwidth SAN or LAN with shared I/O devices and disk arrays; the cluster acts as a single computer attached to the Internet.

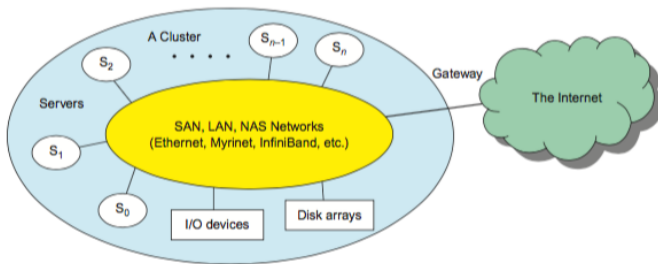
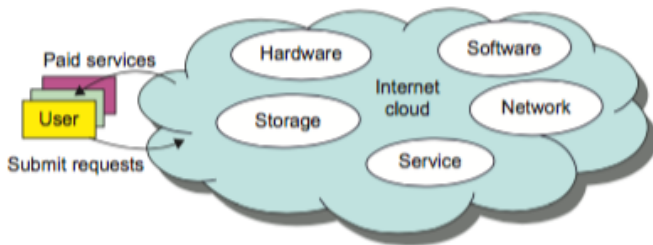


Figure 1.15

Cloud Computing

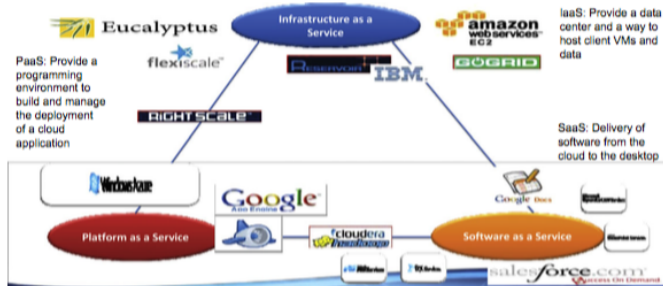
A cloud computing instance

Virtualized resources from data centers to form an Internet cloud, provisioned with hardware, software, storage, network, and services for paid users to run their applications.



The Cloud Landscape

Three cloud service models in a cloud landscape of major providers.



The Cloud Landscape

A stack of six layers of cloud services and their providers.

Cloud application (SaaS)			Concur, RightNOW, Teleo, Kenexa, Webex, Blackbaud, salesforce.com, Netsuite, Kenexa, etc.
Cloud software environment (PaaS)			Force.com, App Engine, Facebook, MS Azure, NetSuite, IBM BlueCloud, SGI Cyclone, eBay
Cloud software infrastructure			Amazon AWS, OpSource Cloud, IBM Ensembles, Rackspace cloud, Windows Azure, HP, Banknorth
Computational resources (IaaS)	Storage (DaaS)	Communications (Caas)	
Collocation cloud services (LaaS)			Savvis, Internap, NTTCommunications, Digital Realty Trust, 365 Main
Network cloud services (NaaS)			Owest, AT&T, AboveNet
Hardware/Virtualization cloud services (HaaS)			VMware, Intel, IBM, XenEnterprise

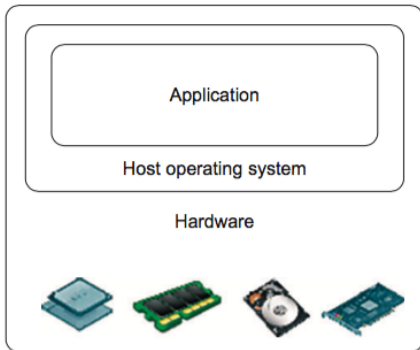
Parallel and Distributed Programming Models

Table 1.7 Parallel and Distributed Programming Models and Tool Sets

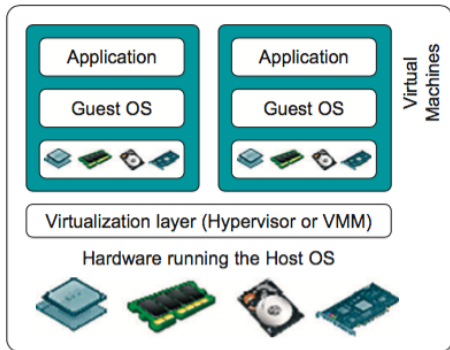
Model	Description	Features
MPI	A library of subprograms that can be called from C or FORTRAN to write parallel programs running on distributed computer systems [6,28,42]	Specify synchronous or asynchronous point-to-point and collective communication commands and I/O operations in user programs for message-passing execution
MapReduce	A web programming model for scalable data processing on large clusters over large data sets, or in web search operations [16]	<i>Map</i> function generates a set of intermediate key/value pairs; <i>Reduce</i> function merges all intermediate values with the same key
Hadoop	A software library to write and run large user applications on vast data sets in business applications (http://hadoop.apache.org/core)	A scalable, economical, efficient, and reliable tool for providing users with easy access of commercial clusters

Virtualization Technologies

Virtualization Technologies



(a) Traditional computer



(b) After virtualization

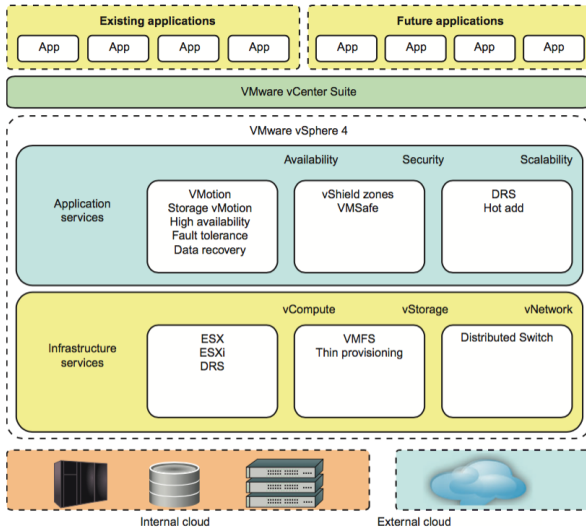
Virtualization Technologies

Table 3.2 Comparison of Four VMM and Hypervisor Software Packages

Provider and References	Host CPU	Host OS	Guest OS	Architecture
VMware Workstation [71]	x86, x86-64	Windows, Linux	Windows, Linux, Solaris, FreeBSD, Netware, OS/2, SCO, BeOS, Darwin	Full Virtualization
VMware ESX Server [71]	x86, x86-64	No host OS	The same as VMware Workstation	Para-Virtualization
Xen [7,13,42]	x86, x86-64, IA-64	NetBSD, Linux, Solaris	FreeBSD, NetBSD, Linux, Solaris, Windows XP and 2003 Server	Hypervisor
KVM [31]	x86, x86-64, IA-64, S390, PowerPC	Linux	Linux, Windows, FreeBSD, Solaris	Para-Virtualization

Virtualization Technologies

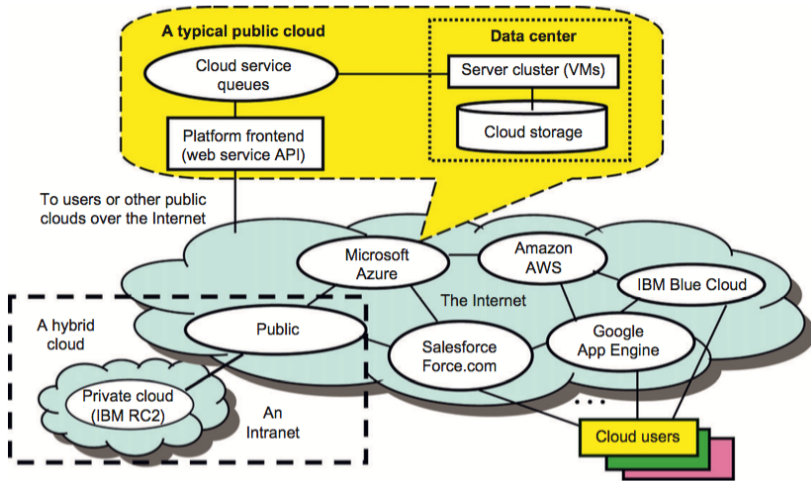
VMware ESX vSphere/4, a cloud operating system that manages compute, storage, and network resources over virtualized data centers.



Cloud Computing Platform

Cloud Computing Platform

Public, private, and hybrid clouds illustrated by functional architecture and connectivity of representative clouds available by 2011.



Cloud Computing Platform

Table 4.1 Public Cloud Offerings of IaaS [10,18]

Cloud Name	VM Instance Capacity	API and Access Tools	Hypervisor, Guest OS
Amazon EC2	Each instance has 1–20 EC2 processors, 1.7–15 GB of memory, and 160–1.69 TB of storage.	CLI or web Service (WS) portal	Xen, Linux, Windows
GoGrid	Each instance has 1–6 CPUs, 0.5–8 GB of memory, and 30–480 GB of storage.	REST, Java, PHP, Python, Ruby	Xen, Linux, Windows
Rackspace Cloud	Each instance has a four-core CPU, 0.25–16 GB of memory, and 10–620 GB of storage.	REST, Python, PHP, Java, C#, .NET	Xen, Linux
FlexiScale in the UK	Each instance has 1–4 CPUs, 0.5–16 GB of memory, and 20–270 GB of storage.	web console	Xen, Linux, Windows
Joyent Cloud	Each instance has up to eight CPUs, 0.25–32 GB of memory, and 30–480 GB of storage.	No specific API, SSH, Virtual/Min	OS-level virtualization, OpenSolaris

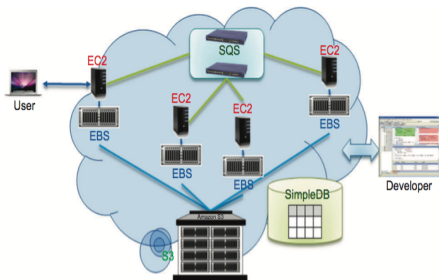
Cloud Computing Platform

Table 4.2 Five Public Cloud Offerings of PaaS [10,18]

Cloud Name	Languages and Developer Tools	Programming Models Supported by Provider	Target Applications and Storage Option
Google App Engine	Python, Java, and Eclipse-based IDE	MapReduce, web programming on demand	Web applications and BigTable storage
Salesforce.com's Force.com	Apex, Eclipse-based IDE, web-based Wizard	Workflow, Excel-like formula, Web programming on demand	Business applications such as CRM
Microsoft Azure	.NET, Azure tools for MS Visual Studio	Unrestricted model	Enterprise and web applications
Amazon Elastic MapReduce	Hive, Pig, Cascading, Java, Ruby, Perl, Python, PHP, R, C++	MapReduce	Data processing and e-commerce
Aneka	.NET, stand-alone SDK	Threads, task, MapReduce	.NET enterprise applications, HPC

Amazon Web Services (AWS)

Amazon Web Services



- ▶ Amazon applies the IaaS model (<http://aws.amazon.com/>).
- ▶ **EC2** provides the virtualized platforms to the host VMs where the cloud application can run.
- ▶ **S3** (Simple Storage Service) provides the object-oriented storage service for users.
- ▶ **EBS** (Elastic Block Service) provides the block storage interface which can be used to support traditional applications.
- ▶ **SQS** stands for Simple Queue Service, and its job is to ensure a reliable message service between two processes.

Amazon Web Services

Table 4.6 AWS Offerings in 2011

Service Area	Service Modules and Abbreviated Names
Compute	Elastic Compute Cloud (EC2), Elastic MapReduce, Auto Scaling
Messaging	Simple Queue Service (SQS), Simple Notification Service (SNS)
Storage	Simple Storage Service (S3), Elastic Block Storage (EBS), AWS Import/Export
Content Delivery	Amazon CloudFront
Monitoring	Amazon CloudWatch
Support	AWS Premium Support
Database	Amazon SimpleDB, Relational Database Service (RDS)
Networking	Virtual Private Cloud (VPC) (Example 4.1, Figure 4.6), Elastic Load Balancing
Web Traffic	Alexa Web Information Service, Alexa Web Sites
E-Commerce	Fulfillment Web Service (FWS)
Payments and Billing	Flexible Payments Service (FPS), Amazon DevPay
Workforce	Amazon Mechanical Turk

(Courtesy of Amazon, <http://aws.amazon.com> [3])

Programming on Amazon EC2

- ▶ Amazon was the first company to introduce VMs in application hosting.
- ▶ Customers can rent VMs instead of physical machines to run their own applications.
- ▶ The elastic feature of such a service is that a customer can create, launch, and terminate server instances as needed, paying by the hour for active servers.
- ▶ Amazon provides several types of pre-installed VMs. Instances are often called **Amazon Machine Images (AMIs)** which are pre-configured with operating systems based on Linux or Windows, and additional software.

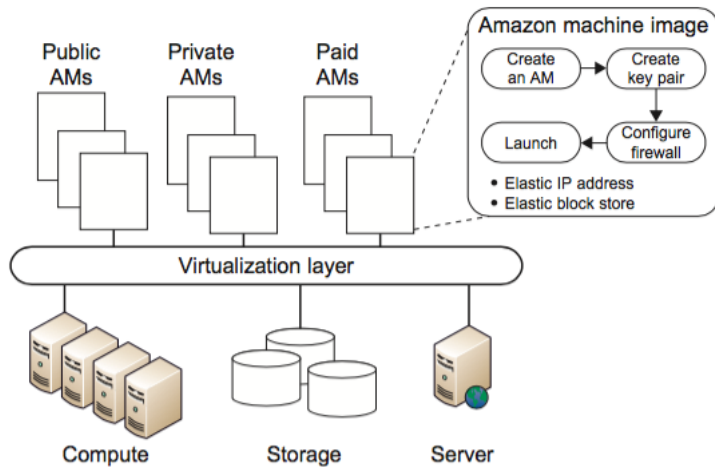
Programming on Amazon EC2

Table 6.12 Three Types of AMI

Image Type	AMI Definition
Private AMI	Images created by you, which are private by default. You can grant access to other users to launch your private images.
Public AMI	Images created by users and released to the AWS community, so anyone can launch instances based on them and use them any way they like. AWS lists all public images at http://developer.amazonwebservices.com/connect/kbcategory.jspa?categoryID=171 .
Paid QAMI	You can create images providing specific functions that can be launched by anyone willing to pay you per each hour of usage on top of Amazon's charges.

Programming on Amazon EC2

Amazon EC2 execution environment.



Programming on Amazon EC2

IaaS instances available in October 2010 in five broad classes:

1. **Standard instances** are well suited for most applications.
2. **Micro instances** provide a small number of consistent CPU resources and allow you to burst CPU capacity when additional cycles are available. They are well suited for lower throughput applications and web sites that consume significant compute cycles periodically.

Programming on Amazon EC2

IaaS instances available in October 2010 in five broad classes:

1. **High-CPU instances** have proportionally more CPU resources than memory (RAM) and are well suited for compute-intensive applications.
2. **Cluster compute instances** provide proportionally high CPU resources with increased network performance and are well suited for high-performance computing (HPC) applications and other demanding network-bound applications. They use 10 Gigabit Ethernet interconnections.

Programming on Amazon EC2

Table 6.13 Instance Types Available on Amazon EC2 (October 6, 2010)

Compute Instance	Memory GB	ECU or EC2 Units	Virtual Cores	Storage GB	32/64 Bit
Standard: small	1.7	1	1	160	32
Standard: large	7.5	4	2	850	64
Standard: extra large	15	8	4	1690	64
Micro	0.613	Up to 2		Only EBS	32 or 64
High-memory	17.1	6.5	2	420	64
High-memory: double	34.2	13	4	850	64
High-memory: quadruple	68.4	26	8	1690	64
High-CPU: medium	1.7	5	2	350	32
High-CPU: extra large	7	20	8	1690	64
Cluster compute	23	33.5	8	1690	64

Programming on Amazon EC2

Table 6.14 Cost of Amazon On-Demand VM Instant Types (October 6, 2010)

VM Instance Type	Size	Linux/UNIX Usage	Windows Usage
Standard instances	Small (default)	\$0.085 per hour	\$0.12 per hour
	Large	\$0.34 per hour	\$0.48 per hour
	Extra large	\$0.68 per hour	\$0.96 per hour
Micro instances	Micro	\$0.02 per hour	\$0.03 per hour
High-memory instances	Extra large	\$0.50 per hour	\$0.62 per hour
	Double extra large	\$1.00 per hour	\$1.24 per hour
	Quadruple extra large	\$2.00 per hour	\$2.48 per hour
Cluster compute instances	Quadruple extra large	\$1.60 per hour	Not available

Programming on Amazon EC2 Instances

vCPU — Memory (GiB)— Storage (GB) — Net. Performance — Physical Processor — Clock Speed (GHz)

- ▶ T2 instances are Burstable Performance Instances that provide a baseline level of CPU performance with the ability to burst above the baseline.

t2.nano	1	0,5	Solo EBS	Basse	Famiglia Intel Xeon	Fino a 3,3
t2.micro	1	1	Solo EBS	Da basso a moderato	Famiglia Intel Xeon	Fino a 3,3
t2.small	1	2	Solo EBS	Da basso a moderato	Famiglia Intel Xeon	Fino a 3,3
t2.medium	2	4	Solo EBS	Da basso a moderato	Famiglia Intel Xeon	Fino a 3,3
t2.large	2	8	Solo EBS	Da basso a moderato	Famiglia Intel Xeon	Fino a 3
t2.xlarge	4	16	Solo EBS	Moderato	Famiglia Intel Xeon	Fino a 3
t2.2xlarge	8	32	Solo EBS	Moderato	Famiglia Intel Xeon	Fino a 3

Programming on Amazon EC2 Instances

vCPU — Memory (GiB)— Storage (GB) — Net. Performance — Physical Processor — Clock Speed (GHz)

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- ▶ M4 instances are the latest generation of General Purpose Instances. This family provides a balance of compute, memory, and network resources, and it is a good choice for many applications.

m4.large	2	8	Solo EBS	Moderato	Intel Xeon E5-2676 v3**	2,4
m4.xlarge	4	16	Solo EBS	Alto	Intel Xeon E5-2676 v3**	2,4
m4.2xlarge	8	32	Solo EBS	Alto	Intel Xeon E5-2676 v3**	2,4
m4.4xlarge	16	64	Solo EBS	Alto	Intel Xeon E5-2676 v3**	2,4
m4.10xlarge	40	160	Solo EBS	10 Gigabit	Intel Xeon E5-2676 v3	2,4
m4.16xlarge	64	256	Solo EBS	20 Gigabit	Intel Xeon E5-2686 v4	2,3

Programming on Amazon EC2 Instances

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- ▶ This family includes the M3 instance types and provides a balance of compute, memory, and network resources, and it is a good choice for many applications.

m3.medium	1	3,75	1 x 4 SSD	Moderato	Intel Xeon E5-2670 v2*	2,5
m3.large	2	7,5	1 x 32 SSD	Moderato	Intel Xeon E5-2670 v2*	2,5
m3.xlarge	4	15	2 x 40 SSD	Alto	Intel Xeon E5-2670 v2*	2,5
m3.2xlarge	8	30	2 x 80 SSD	Alto	Intel Xeon E5-2670 v2*	2,5

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- ▶ C4 instances are the latest generation of Compute-optimized instances, featuring the highest performing processors and the lowest price/compute performance in EC2.

c4.large	2	3,75	Solo EBS	Moderato	Intel Xeon E5-2666 v3	2,9
c4.xlarge	4	7,5	Solo EBS	Alto	Intel Xeon E5-2666 v3	2,9
c4.2xlarge	8	15	Solo EBS	Alto	Intel Xeon E5-2666 v3	2,9
c4.4xlarge	16	30	Solo EBS	Alto	Intel Xeon E5-2666 v3	2,9
c4.8xlarge	36	60	Solo EBS	10 Gigabit	Intel Xeon E5-2666 v3	2,9

Programming on Amazon EC2 Instances

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- ▶ C3.

c3.large	2	3,75	2 x 16 SSD	Moderato	Intel Xeon E5-2680 v2	2,8
c3.xlarge	4	7,5	2 x 40 SSD	Moderato	Intel Xeon E5-2680 v2	2,8
c3.2xlarge	8	15	2 x 80 SSD	Alto	Intel Xeon E5-2680 v2	2,8
c3.4xlarge	16	30	2 x 160 SSD	Alto	Intel Xeon E5-2680 v2	2,8
c3.8xlarge	32	60	2 x 320 SSD	10 Gigabit	Intel Xeon E5-2680 v2	2,8

Programming on Amazon EC2 Instances

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- ▶ C3.
- ▶ X1 Instances are optimized for large-scale, enterprise-class, in-memory applications and have the lowest price per GiB of RAM among Amazon EC2 instance types.

x1.16large	64	976	1 x 1.920 SSD	10 Gigabit	Intel Xeon E7-8880 v3	2,3
x1.32xlarge	128	1.952	2 x 1.920 SSD	20 Gigabit	Intel Xeon E7-8880 v3	2,3

Programming on Amazon EC2 Instances

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- ▶ C3.
- ▶ X1 Instances are optimized for large-scale, enterprise-class, in-memory applications and have the lowest price per GiB of RAM among Amazon EC2 instance types.
- ▶ R4 instances are optimized for memory-intensive applications and offer better price per GiB of RAM than R3.

r4.large	2	15,25	-	Fino a 10 Gigabit	Intel Xeon E5-2686 v4	2,3
r4.xlarge	4	30,5	-	Fino a 10 Gigabit	Intel Xeon E5-2686 v4	2,3
r4.2xlarge	8	61	-	Fino a 10 Gigabit	Intel Xeon E5-2686 v4	2,3
r4.4xlarge	16	122	-	Fino a 10 Gigabit	Intel Xeon E5-2686 v4	2,3
r4.8xlarge	32	244	-	10 Gigabit	Intel Xeon E5-2686 v4	2,3
r4.16xlarge	64	488	-	20 Gigabit	Intel Xeon E5-2686 v4	2,3

Programming on Amazon EC2 Instances

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- ▶ R3 instances are optimized for memory-intensive applications and offer lower price per GiB of RAM.

r3.large	2	15,25	1 x 32 SSD	Moderato	Intel Xeon E5-2670 v2	2,5
r3.xlarge	4	30,5	1 x 80 SSD	Moderato	Intel Xeon E5-2670 v2	2,5
r3.2xlarge	8	61	1 x 160 SSD	Alto	Intel Xeon E5-2670 v2	2,5
r3.4xlarge	16	122	1 x 320 SSD	Alto	Intel Xeon E5-2670 v2	2,5
r3.8xlarge	32	244	2 x 320 SSD	10 Gigabit	Intel Xeon E5-2670 v2	2,5

Programming on Amazon EC2 Instances

vCPU — Memory (GiB)— Storage (GB) — Net. Performance — Physical Processor — Clock Speed (GHz)

- ▶ P2 instances are intended for general-purpose GPU compute applications.

p2.xlarge	4	61	Solo EBS	Alto	Intel Xeon E5-2686 v4	2,3
p2.8xlarge	32	488	Solo EBS	10 Gigabit	Intel Xeon E5-2686 v4	2,3
p2.16xlarge	64	732	Solo EBS	20 Gigabit	Intel Xeon E5-2686 v4	2,3

Programming on Amazon EC2 Instances

vCPU — Memory (GiB)— Storage (GB) — Net. Performance — Physical Processor — Clock Speed (GHz)

- ▶ P2 instances are intended for general-purpose GPU compute applications.
- ▶ G2 instances are optimized for graphics-intensive applications.

g2.2xlarge	8	15	1 x 60 SSD	Alto	Intel Xeon E5-2670 v*	2,6
g2.8xlarge	32	60	2 x 120 SSD	10 Gigabit	Intel Xeon E5-2670	2,6

Programming on Amazon EC2 Instances

vCPU — Memory (GiB)— Storage (GB) — Net. Performance — Physical Processor — Clock Speed (GHz)

- ▶ P2 instances are intended for general-purpose GPU compute applications.
- ▶ G2 instances are optimized for graphics-intensive applications.
- ▶ F1 instances offer customizable hardware acceleration with field programmable gate arrays (FPGAs).

f1.2xlarge	8	122	480 SSD	High	Intel Xeon E5-2686 v4	2,3
f1.16xlarge	64	976	4 x 960	High	Intel Xeon E5-2686 v4	2,3

Programming on Amazon EC2 Instances

vCPU — Memory (GiB)— Storage (GB) — Net. Performance — Physical Processor — Clock Speed (GHz)

- ▶ P2 instances are intended for general-purpose GPU compute applications.
- ▶ G2 instances are optimized for graphics-intensive applications.
- ▶ F1 instances offer customizable hardware acceleration with field programmable gate arrays (FPGAs).
- ▶ This family includes the High Storage Instances that provide Non-Volatile Memory Express (NVMe) SSD backed instance storage optimized for low latency, very high random I/O performance, high sequential read throughput and provide high IOPS at a low cost.

i3.large	2	15,25	1 x 475 SSD NVMe	Fino a 10 Gigabit	Intel Xeon E5-2686 v4	2,3
i3.xlarge	4	30,5	1 x 950 SSD NVMe	Fino a 10 Gigabit	Intel Xeon E5-2686 v4	2,3
i3.2xlarge	8	61	1 x 1.900 SSD NVMe	Fino a 10 Gigabit	Intel Xeon E5-2686 v4	2,3
i3.4xlarge	16	122	2 x 1.900 SSD NVMe	Fino a 10 Gigabit	Intel Xeon E5-2686 v4	2,3
i3.8xlarge	32	244	4 x 1.900 SSD NVMe	10 Gigabit	Intel Xeon E5-2686 v4	2,3
i3.16large	64	488	8 x 1.900 SSD NVMe	20 Gigabit	Intel Xeon E5-2686 v4	2,3

Programming on Amazon EC2 Instances

vCPU — Memory (GiB)— Storage (GB) — Net. Performance — Physical Processor — Clock Speed (GHz)

- ▶ P2 instances are intended for general-purpose GPU compute applications.
- ▶ G2 instances are optimized for graphics-intensive applications.
- ▶ F1 instances offer customizable hardware acceleration with field programmable gate arrays (FPGAs).
- ▶ This family includes the High Storage Instances that provide Non-Volatile Memory Express (NVMe) SSD backed instance storage optimized for low latency, very high random I/O performance, high sequential read throughput and provide high IOPS at a low cost.
- ▶ D2 instances feature up to 48 TB of HDD-based local storage, deliver high disk throughput, and offer the lowest price per disk throughput performance on Amazon EC2.

d2.xlarge	4	30,5	3 x 2000	Moderato	Intel Xeon E5-2676 v3	2,4
d2.2xlarge	8	61	6 x 2000	Alto	Intel Xeon E5-2676 v3	2,4
d2.4xlarge	16	122	12 x 2000	Alto	Intel Xeon E5-2676 v3	2,4
d2.8xlarge	36	244	24 x 2000	10 Gigabit	Intel Xeon E5-2676 v3	2,4

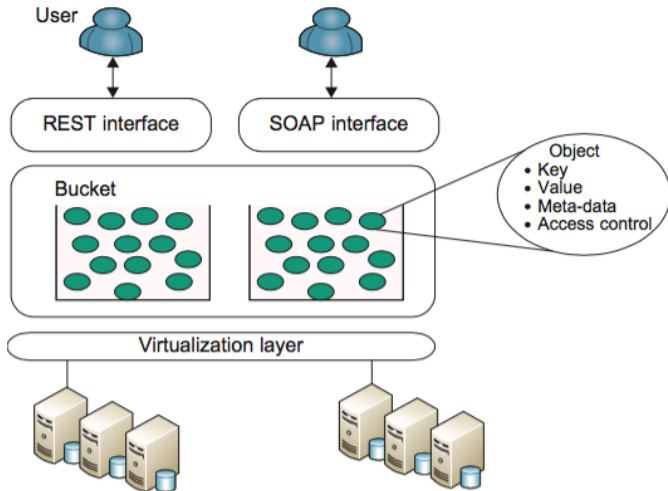
Programming on Amazon EC2

Amazon S3 execution environment.

- ▶ Amazon **S3** provides a simple web services interface that can be used to store and retrieve any amount of data, at any time, from anywhere on the web.
- ▶ S3 provides the object-oriented storage service for users.
- ▶ Users can access their objects through Simple Object Access Protocol (SOAP) or REpresentational State Transfer (REST).

Programming on Amazon EC2

Amazon S3 execution environment.



Programming on Amazon EC2

Amazon Elastic Block Store (EBS) and SimpleDB.

- ▶ **Elastic Block Store (EBS)** provides the volume block interface for saving and restoring the virtual images of EC2 instances.
- ▶ S3 is “Storage as a Service” with a messaging interface. EBS is analogous to a distributed file system accessed by traditional OS disk access mechanisms.
- ▶ EBS allows you to create storage volumes from 1 GB to 1 TB that can be mounted as EC2 instances.
- ▶ Multiple volumes can be mounted to the same instance.
- ▶ In terms of pricing, Amazon provides a similar pay-per-use schema as EC2 and S3.

Programming on Amazon EC2

Amazon Elastic Block Store (EBS) and SimpleDB.

- ▶ **SimpleDB** provides a simplified data model based on the relational database data model.
- ▶ Structured data from users must be organized into domains. Each domain can be considered a table.
- ▶ Items are the rows in the table. A cell in the table is recognized as the value for a specific attribute (column name) of the corresponding row.
- ▶ In this model is possible to assign multiple values to a single cell in the table. This is not permitted in a traditional relational database which wants to maintain data consistency.

Programming on Amazon EC2

Others Technologies 2017.

- ▶ **Amazon DynamoDB** – database NoSQL.
- ▶ **Amazon Cognito** – Authentication service, allows to easily manage users connection also by for instance Facebook, Twitter and Amazon.
- ▶ **AWS CodeCommit** – source control based on Git protocol.
- ▶ **Amazon CloudWatch** – active monitoring of AWS resources and applications running on AWS.
- ▶ **AWS X-Ray** – analysis and debugging of production and distributed applications.
- ▶ **Amazon Mobile Analytics** – measuring usage and revenue from app.
- ▶ **Amazon Chime** – a communications service that transforms online meetings with a secure, easy-to-use application that you can trust.

Programming on Amazon EC2

Others Technologies 2017.

- ▶ **Amazon Simple Workflow Service (SWF)** – makes it easy to build applications that use Amazon's cloud to coordinate work across distributed components. In Amazon SWF, a task represents a logical unit of work that is performed by a component of your workflow. Coordinating tasks in a workflow involves managing intertask dependencies, scheduling, and concurrency in accordance with the logical flow of the application.
- ▶ **Amazon Simple Queue Service (SQS)** e **Amazon Simple Notification Service (SNS)**.
- ▶ **Amazon Elastic Transcoder** – lets you convert media files that you have stored in Amazon S3 into media files in the formats required by consumer playback devices.
- ▶ **AWS Key Management Service** – create and control the keys used to encrypt your data.

Programming on Amazon EC2

Others Technologies 2017.

- ▶ **AWS Key Management Service** – create and control the keys used to encrypt your data.
- ▶ **AWS Lambda** – a compute service that lets you run code without provisioning or managing servers:
 - ▶ executes your code only when needed and scales automatically, from a few requests per day to thousands per second;
 - ▶ you pay only for the compute time you consume;
 - ▶ you can run code for virtually any type of application or backend service - all with zero administration;
 - ▶ write in any languages that AWS Lambda supports (currently Node.js, Java, C# and Python).

Programming on Amazon EC2

Others Technologies 2017.

- ▶ **AWS CodePipeline** – a continuous delivery service you can use to model, visualize, and automate the steps required to release your software.
- ▶ **AWS Device Farm** – an app testing service that enables you to test and interact with your Android, iOS, and Web apps on real, physical phones and tablets that are hosted by Amazon Web Services (AWS).
- ▶ **AWS Step Functions** – a new service that launched at 2016, makes it easier to build complex, distributed applications in the cloud:
 - ▶ this service allow to create state machines that can connect microservices and activities into a visual workflow;
 - ▶ the state machines support branching, parallel execution, retry/error handling, synchronization (via Wait states), and task execution (via AWS Lambda or an AWS Step Functions Activity).

Programming on Amazon EC2

Others Technologies 2017.

- ▶ **Amazon Simple Email Service SES.**
- ▶ **Amazon QuickSight** – a fast business analytics service to build Visualizations, perform ad hoc analysis, and quickly get business insights from your data.
- ▶ **AWS CodeBuild** – compile and run the code test with continuous resizing. Paying only for the compilation time.

Benchmarking

What is Scalability?

- ▶ Scalability measures the capability of the system to react to the computational resources used.
- ▶ Scalability can be seen in a lot of forms – speed, efficiency, high reliable applications and heterogeneity:
 1. *Load scalability.* A system is load scalable if it is able to exploits the available resources in heavy loads conditions.
 2. *Space scalability.* A system is space scalable when it is able to maintain the memory requirements under some reasonable levels (also when the size of the input is large).

What is Scalability?

1. *Space-time scalability.* A system is space-time scalable when it provides the same performance whether the system is of moderate size or large.
2. *Structural scalability.* A system is structurally scalable if its implementation or architecture does not limit the number of resources or data input.

When a system is scalable?

- ▶ The scalability of a system can be analyzed according to different properties:
 - ▶ *speed*, that is increasing the number of processing resources provides an increasing speed;
 - ▶ *efficiency*, the efficiency remain unchanged when the processors and the problem size increases;
 - ▶ *size*, the maximum number of computational resources that a system can accommodate;
 - ▶ *application*, a software is scalable when it provides better performance when it is executed on a larger system;
 - ▶ *generation*, a scalable system should achieve better performance according to the generation of components used;
 - ▶ *heterogeneous*, a scalable system should be able to exploit different hardware and software components.

How to Measure Scalability Efficiency

There are two basic ways to measure the parallel performance of a given system, depending on whether or not one is cpu-bound or memory-bound.

- ▶ *Strong Scalability.*
- ▶ *Weak Scalability.*

How to Measure Scalability Efficiency

Strong Scalability (cpu-bound)

- ▶ The idea is to study the number of resources needed, to a given application in order to complete the computation in a reasonable time.
- ▶ For this reason the problem size stays fixed but the number of processing elements are increased.
- ▶ An application scales linearly when the speedup obtained is equal to the computational resources used n .

How to Measure Scalability Efficiency

Weak Scalability (memory-bound)

- ▶ The idea is to define the efficiency of an application fixing the problem size for each computational resource.
- ▶ This measure is useful for studying the memory or resources consumption of an application.
- ▶ The linear scaling is achieved if the run time stays constant while the workload is increased in direct proportion to the number of computational resources

Example

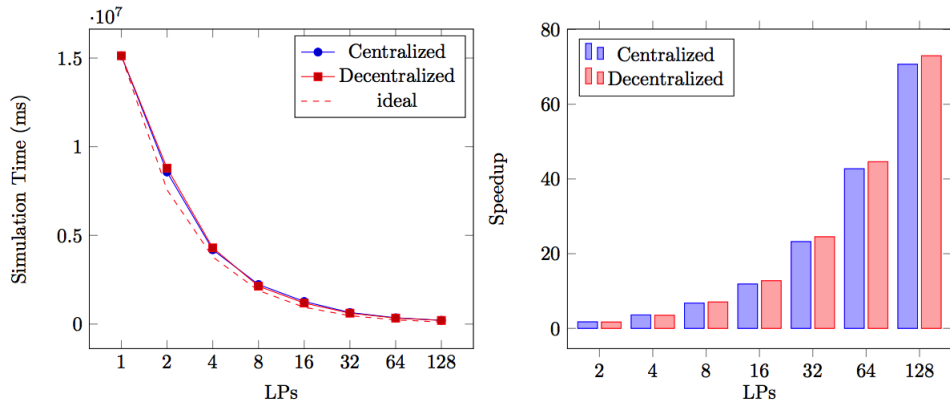


Figure 7: Strong Scalability of Woims 3D simulation.

Thank you for your attention . . .

Next Lesson Amazon Web Services: Management



cspagnuolo@unisa.it

<http://www.carminespagnuolo.eu/>